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THE EFFECTS OF PSYCHOLOGICAL STRESS UPON PERFORMANCE¹

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An understanding of the effect of psychological stress upon skilled performance is of great theoretical and practical importance. People are often faced with the necessity of performing skilled work under conditions which are highly stressful. Such is obviously the case in military combat. The effectiveness of a pilot, gunner, or radar observer must be maintained even when he is threatened by physical injury or harassed by the need to hurry the performance of a complicated task. The obvious fact that human beings are often required to work under stress does not call for further elaboration.

The problems of stress involve questions of emotions, motivation, and learning. Many theoretical issues in these fields are of basic importance in an analysis of the effects of stress upon performance. This fact has been recognized; most of the experimental work upon stress has been undertaken for theoretical rather than practical reasons. The problem of the effects of stress cuts across many fields. We are able to draw many hypotheses concerning stress from the theoretical constructs of motivation, emotion, and learning.

We shall begin the discussion of the work in this field with an analysis of the concept of stress. Following this we shall describe the various experimental techniques used to induce stress. This aspect of the problem is important because there are many methodological difficulties inherent in the production of a genuinely stressful situation. Then we shall review, briefly, the kinds of performance which have been studied under stress. Finally, we shall discuss some of the theoretical implications of the work on stress and skilled performance.

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THE CONCEPT OF STRESS

It is not possible to discuss intelligently the work on psychological stress without dealing with the problem of the concept of stress. The definitions of stress that have been given from time to time are inadequate for several reasons. It is possible to think of stress in terms of situations. For example, we say that a crucial examination is stressful to the participants, or that combat is stressful to soldiers. One difficulty with this approach is that these situations are not reacted to uniformly by all people. We cannot predict the behavior of individuals by simply describing the situation. One person may tremble, sweat, experience discomfort, and show signs of behavioral disorganization. Another may show an impairment in performance with no other subjective concomitants. Still others may show no measurable effects from the situation.

In most of the research on stress, the experimenter selects a situation which, from past experience, seems to be threatening to most people. Implicit in this selection is the necessity of identifying stress with the motivations of the people who are being tested. However, because people differ in motivations and in the ways they deal with them, it is never really possible to define a general stress situation. The situation will be more or less stressful for the individual members of the group, and it is likely that these differences in the meaning of the situation will appear in terms of performance.

It is also possible to define stress by emphasizing the reactions or responses of an individual rather than the situation. The trouble with this approach is similar to that encountered when we emphasize the situation. What kinds of reactions should we measure? Are we to consider changes in skilled performance as the "measure" of stress, or are we to consider changes in subjective report? It is apparent that these things may change independently of one another. Moreover, these changes are a function of many unrelated variables. For example, skilled performance may be affected by a change in motivation or a change in approach to the task. It would be meaningless to identify these changes as the effects of stress.

Since stress cannot be defined in terms of stimulus or response operations alone, it is necessary to think of it in terms of an intervening variable.² The additional concept that is necessary is that of motiva-

² Since the submission of this manuscript an interesting article by Brown and Farber, entitled "Emotions conceptualized as intervening variables with suggestions toward a theory of frustration," has appeared in this JOURNAL (November 1951, 48, 465-495). In it Brown and Farber offer some opinions which appear to be parallel with, and certainly related to, our theoretical discussion of psychological stress.

tion. Stress, therefore, is really a secondary concept, built upon the relationship between a primary concept, motivation, and the situation in which motivated behavior appears. We would then think that stress occurs when a particular situation threatens the attainment of some goal. The actual responses that the individual may show will depend partly upon the kinds of mechanisms that have been previously established.

This viewpoint demands that the concept of motivation itself be explored. The psychologist who is interested in problems of human behavior finds it very difficult to estimate from measures of behavior the kind and degree of motivation involved in a particular situation. There is general agreement among psychologists that it is ultimately essential to do this in order to account for the enormous individual differences that are found in behavior. In studies of psychological stress individual differences tend to be one of the main findings.

Pointing out the parallel between the problems of psychological stress and those of physiological stress will illustrate the difficulty in dealing with motivation. Physiological stress does not seem to involve the same definitional problems that psychological stress does, because the "motivational component" in physiological stress is stated in terms of the well-worked-out mechanisms of homeostasis. Selye (34) has defined physiological stress as any condition that produces the "adaptation-syndrome," which is the reaction of the organism in returning to the homeostatic state.

The psychologist has no adequate way of defining the psychological condition that corresponds to the homeostatic steady state. Consequently, the use of the term *stress* must necessarily be a little looser than we would like it to be. When we speak of tension-systems, what we are really doing is postulating a psychological steady-state as a lack of tension. What needs to be investigated are the properties of such a state and deviations from it.

The solution of most experimenters who have studied the responses of groups under stress has been to produce situations which are thought to thwart the motives of most people. This is an adequate solution as long as one is not attempting to account for the reactions of any individual. If the experimenter tries to account for a particular individual's response, this assumption is not satisfactory.

EXPERIMENTAL PROCEDURES FOR PRODUCING STRESS

The principal problem in the study of behavior under stress has been the production of realistic stress situations. A variety of techniques have been tried. Indeed, it might be said that no two experimental

studies in the literature exactly duplicate the same technique. This variety of method has led to considerable confusion, since it is likely that each of these techniques has a somewhat different effect upon performance. It is important to review the main techniques that have been used. These techniques fall into two main classes: (a) stress induced through failure, and (b) stress induced by the task itself.

Stress induced by failure. Failure or threat of failure at a task has been the method most frequently used in experiments on stress. This has been specifically done in the following ways:

1. By presenting the subject with an unsolvable task. In this procedure the subject must work at a task that, unknown to him, cannot be solved. There are a number of ways in which this condition may be established. For example, anagrams for which there are no solutions may be mixed with a group of solvable ones. Or, in another case, a subject may be required to retain a list of digits which is beyond any individual's memory span.

2. In another type of failure-stress, the subject may be interrupted at the task before he could possibly have finished. The task may consist of a group of arithmetic problems. Before he has finished all of the problems, he is interrupted, and told that his time is up.

3. A common technique for the production of failure-stress is the introduction of false norms which indicate failure even if performance has been adequate. The use of this technique may incorporate aspects of the first two procedures. For example, an individual may be interrupted before he has completed all of the items on an arithmetic test, and then told by the experimenter that anyone who has not finished has done poorly. The subject may be told at the outset of the experiment that anyone with normal intelligence (or anyone who expects to be successful in college, or have a decent career in the Air Force, etc.) ought to be able to complete this task within the time limit. In the latter case, failure to complete the problems automatically tells the subject that he is doing poorly.

These techniques of failure-stress present some difficulties. One of the most important experimental limitations is the problem of the control of the subject's motivation. In order for a failure situation to be stressful it is necessary for the individual to be motivated to succeed, or at least to avoid failure. While most subjects are anxious to succeed, the real problem is that all subjects are certainly not equally anxious. Some may not be motivated at all. The usual assumption is that enough of the subjects will be sufficiently involved to become stressed. Some subjects will be seriously disturbed by the threat of failure, while others may be scarcely threatened at all. Of course, the effect of stress will depend upon what the individual expects or demands of himself.

Unfortunately, there is no way of assessing with any confidence the degree of motivation of subjects. We cannot take the effect of stress on performance itself as an indicator of the intensity of motivation because the amount of motivation is not the only determiner of the extent and direction of the reaction to stress. It is certainly to be expected that in some instances strong motivation may produce better performance. On the other hand, it is also capable of producing impaired performance. For the moment, the question of motivation in failure-stress experiments is an important, unsolved problem.

A second difficulty in the failure-stress procedures is presented by the degree of realism that can be produced. In order to be genuinely stressed, subjects must be convinced that the instructions and information given by the experimenter are *bona fide*. Usually the experimenter trusts to his ingenuity and assumes that his instructions will be accepted at face value. There are very few experiments which attempt to assess, directly or indirectly, the degree to which the subjects accept the situation as genuine. There can be no doubt that in most of the experimental situations, the subjects' reactions to the experiment vary from skepticism to severe ego-involvement.

Finally, a seldom considered difficulty in the use of failure-scores to produce the stress is the extent to which a subject will use this false information to alter his mode of attack on the task at hand. Changes in behavior toward the task as a result of failure-information may sometimes have little or nothing to do with stress itself. A subject who has been doing an excellent job may be encouraged (particularly if he is adaptable) to give up a fruitful mode of attack on the grounds that it has been unsuccessful. Thorndike pointed out long ago that an experience of failure may make a subject more variable in performance on the basis of the information it gives him.

Stress induced by working conditions and the task itself. In addition to the experiments in which stress is produced through failure, pressure on the subject may be induced by manipulating the situation in various ways so as to produce excessive demands upon him. Various forms of distractions may be included in this category. Almost any strong sensory input which is extraneous to the task at hand may serve as a source of distraction. In an experimental situation it may be an electric shock, noises, or flashing lights. Distraction can also be produced by verbal disparagement of the subject's performance by the experimenter. Such disparagement may serve as a failure-stress as well as a distraction. In some experiments it is impossible to assess the role of verbal disparagement because it is introduced while the subject is performing, and is therefore both a threat of failure and a distraction.

We have suggested that the manipulation of the experimental situation along the lines of distractions may make difficult demands on the subject, and that these demands then serve as sources of stress. Moreover, some tasks themselves appear to be inherently stressful for much the same reasons. The task may require the subject to attend to too many things at a time, or to perform too many operations at once. These demands may be increased by the experimenter as in the case of rapidly pacing the subject. In some ways this type of stress is like the situation a person faces when he is learning to drive a car. The stress here is probably less a matter of failure—although in some instances it could be—and more of making too many demands at once upon the learner. In other cases there is the added possibility of the threat of personal injury or damage to the car. We would class this stress situation as one induced by the working conditions and the task itself.

We have contrasted these two basic types of stress situations, failure-stress and task-induced stress, because the problem of motivation appears to be a little different for each of them. In the task-induced stress, the motivation depends primarily upon how the subject interprets the experiment. If he sees the distraction or pacing as "something to beat," his effort level is apt to be raised. Motivation can be roughly equated for all subjects by carefully wording the initial instructions, or can be left to vary from subject to subject by ambiguous information. Consider, for example, the effect of instructions which indicate that the purpose of the experiment is to see how well the subject can hold up under distraction as compared with the introduction of distraction as a natural part of the experiment or the task so that the subjects will not react to it as something to be overcome. Despite the methodological problems raised here, the problem for the experimenter is simpler in the task-induced stress than it is in the failure-stress experiment. The failure-stress situations depend primarily on the production of a realistic threat to the subjects' self esteem or to some goal-oriented behavior. The variability of reactions to this kind of condition must be considerably greater than to the situation where stress is induced by manipulation of the working conditions or by the excessive demands of the task.

Evaluating the differences in technique. Unfortunately, we can only guess, at present, what the effects of the differences in the techniques of production of stress might be. Most of the experimental situations which have been used in the study of stress are not directly comparable with one another. It is almost impossible to generalize with confidence from the study of the effects of any one experimental stress-producing technique. One of the important problems for future research is the

study of the differences in the effects of the various procedures for inducing stress.

THE MEASUREMENT OF THE EFFECTS OF STRESS

One of the chief difficulties in the design of stress experiments is concerned with the measurement of the effects of stress upon performance. It is important to measure the effects of stress unconfounded with any other variable. The problem arises because it is often very difficult to get a measure of reaction to stress which is independent of the subject's ability to learn or perform certain kinds of tasks.

For example, if the experimental problem is to study the effects of some kind of stress upon the learning of a psychomotor skill, the question arises, "How can we differentiate the subject's 'normal' rate of learning from his rate of learning as it is affected by the stress condition?" For data which deal with groups of subjects there is a simple solution to the difficulty. It is possible, through control subjects, to estimate the typical rate of learning and compare it with the rate of learning of a comparable group of subjects under some stress condition. If the experiment is properly designed, and the control group appropriately selected so that it is adequately matched with the experimental group, then we can observe the general effects of stress as a difference in the mean performance or a difference in the variability of performance between the two groups.

The tricky problem actually arises when we are concerned with the performance of a particular individual under stress. Whenever we are engaged in predicting performance under stress from personality measures, it is necessary to obtain a measure for that particular subject. This "stress score" must be measured independently of the subject's initial ability and his change in performance due to fatigue or learning. Because individuals differ so much with respect to their basic abilities and rates of change, some estimate of each subject's performance without stress is necessary before correlational techniques may be used.

It has been possible to solve this dilemma in three ways. The most obvious technique has been to test each subject twice—a pretest under standard conditions followed by a stress test. In that way, a difference score may be obtained between performance under stress and performance without stress. If this difference score correlates with the score on the first test (as it usually does) then it is necessary to correct this difference measure so that it is free from the subject's level of performance on the first test. It is a statistical fact that subjects who do well on the first test tend to go down on retest, while subjects who do badly

on the first test improve on the second test. This means that difference scores cannot be used to represent stress scores without a correction for regression on the first test. When this correction has been made, the resulting score represents the effect of stress upon performance for any one subject.

Another technique for obtaining individual stress scores is to make use of plateaus in the learning curves, or to select tasks for study that show little or no change in performance with training. In the case of plateaus, if a subject has attained a point in the learning of a skill where little or no further change in performance is expected for some time, this level might be used as a reference point against which to compare his performance under stressful conditions. Or the plateau may be induced by giving a subject so much practice that he attains his maximum achievement on the task, and no further changes are expected.

The trouble with plateaus is that they are seldom very dependable. Most plateaus are illusory, or cannot be depended upon to remain steady long enough to introduce the stress condition satisfactorily. In the case of the final plateau, where subjects have been practiced to a maximum of performance, this level is often really not a true maximum, and further practice produces a spurt in performance. Furthermore, this latter technique, used by Williams (43), has the disadvantage of producing a stress situation which can only result in a decrement in performance if the subjects have reached their physiological limit.

In summary, it may be said that it is necessary to provide some kind of base line or standard of comparison against which to evaluate the uncontaminated effects of stress upon performance. This reference point must be made free of the influence of the subjects' abilities in the skill in question. To do this, the technique which appears to present least difficulty makes use of two test periods with each subject, one under stress and the other under control or standard conditions. We should point out that even this technique is by no means ideal. Difference scores suffer from the difficulty of requiring two equivalent tests for the same subject. Moreover, it may be important which is given first, the control or the stress condition.

A third approach is possible, but so impractical that we shall dismiss it quickly. If it were possible to match individuals beforehand on ability to learn or perform the skill, then one of the pair could be given to the control condition and the other to the experimental condition. Pretests actually could be used to match subjects. However, this technique has not been used because it would be necessary to match not only initial score, but also rate of learning, approach to the task, motiva-

tion, etc. And besides, a good matching is practically unattainable, because there are a great number of uncontrolled variables.

EXPERIMENTAL STUDIES OF PERFORMANCE UNDER STRESS

For convenience, the experimental studies have been classified into those which deal with verbal tasks and those which deal with perceptual-motor tasks. This division is somewhat arbitrary, but follows along more or less traditional lines. It is possible that a more rational classification may arise out of further experimental work.

The effects of stress upon verbal performance. Several experimenters have studied the effects of stress upon intelligence, primarily verbal, in children. Using nine-year-old boys, Lantz (15) obtained a statistically significant impairment of Stanford-Binet scores following a failure experience, but no such effect after a successful experience. An examination by Lantz of the differential effects of this failure experience upon the various subtests indicated that tasks requiring visual or rote memory were not affected, while those involving reasoning or thinking suffered a decrement.

Along similar lines, an interesting experiment by Hutt (13) differentiated between the effects of failure-stress upon a group of maladjusted children and a group of well-adjusted children. Two methods of administration of the Stanford-Binet were used: (a) an adaptive procedure, in which a failed item was followed by one upon which success was likely, and (b) the standard method, which begins with easy items and ends with items which will be failed. The adaptive method, which produced greater psychological support, resulted in higher average IQ's for the maladjusted group. No difference between the two testing methods was found for the well-adjusted group. This study has considerable implication for the problem of stress in ordinary psychometric procedures.

The great majority of the studies of stress and verbal performance show deterioration or impairment as the result of the experimental conditions. Alper (2) finds a decrement in production in a sentence-formation task as the result of failure-stress. In another experiment, stress induced by pacing and the inherent difficulty of the arithmetic problems produced a reduced rate of learning (40); pacing, in this case, produced failure in 19 out of every 20 trials. Impairment of digit span was obtained with fifth- and sixth-grade children when the number of digits presented was beyond the normal span for these children (44).

Zeller (45), working with nonsense syllables, found a decrement in recall and relearning following an experience of failure; he accounted for

this decrement on the basis of repression of the items which subjects associated with failure. His evidence for this hypothesis was that re-learning and recall improved when the knowledge of failure was removed. Zeller's study may be interpreted as basically a stress experiment. Zeller comments that the subjects' performance progressively deteriorated during the accumulation of failure.

In another failure-stress experiment in which nonsense syllables were learned and recalled, Sullivan (37) found that (a) success produced the most rapid learning and the most complete recall, (b) slowest learning and poorest recall followed failure, and (c) failure was more harmful to a superior group in producing impairment than success was facilitating. The reverse was true for an intellectually inferior group.

Impairment of performance on the digit-symbol test was found by Williams (43) with the use of false failure scores. Williams was primarily interested in validating the Rorschach test. He concentrated upon the prediction of performance under stress by selected Rorschach indices.

In another experiment, stress, induced by failure to complete a task within a specified time limit, produced an increase in errors and variability in time scores of multiplication problems and in the learning of nonsense syllables (21).

Postman and Bruner (30) studied recognition thresholds under stress using three-word sentences. Psychological stress produced by failure and ridicule resulted in poorer performance. Since the stress was administered during actual performance, the results may have been due, in large part, to distraction. The authors observed that during failure, the pre-recognition guesses of the subjects became extremely reckless.

Despite this array of studies showing impairment, poorer performance has not always been found to be the principal effect of stress. Initial decrement but later improvement in code-learning was obtained as the result of stress due to pacing.³ Hurlock (12), on the other hand, found that failure results in an initial improvement and then a later decrement. Lazarus and Eriksen (16) found that the major effect of failure-stress upon the digit symbol test was to produce greater variability between subjects; some subjects improved and others showed impairment.

Finally we might note that there have been a large number of studies aimed at the analysis of selective recall or repression that have some bearing on the problem of the effects of stress (2, 6, 10, 14, 24, 31, 45).

³ DEESE, J., & BOWEN, H. M. The effect of task-induced stress on code learning. Unpublished research, The Johns Hopkins University, 1950.

Eriksen's data (6) suggest that poor or good recall of incompleting tasks may be a function of personality variables related to defense mechanisms.

The effects of stress upon perceptual-motor performance. The general picture of the effects of stress upon perceptual-motor performance is similar to the pattern found with verbal tasks. A large number of studies have shown impairment of perceptual-motor performance under stress. Marquart (23), with the use of visual discrimination, concluded that frustration resulted in slow learning, increased rigidity, and non-adaptive behavior. McClelland and Apicella (20) used card sorting as the experimental task and found that stress induced by false failure-scores resulted in more trials before the criterion was reached. Longer reaction times for completing pictures that were flashed on a screen were obtained by Verville (41) when subjects were given unsolvable tasks prior to the measurement of reaction time. Moreover, slower reaction times were also found by Verville for subjects who were required to solve a number of difficult problems simultaneously before the test of reaction time.

Bayton and Whyte (4), who used a rate of manipulation test, found that a success-failure sequence produced poorer performance than a sequence of failure followed by success. Because they failed to include a control group, it is impossible to decide whether the differences in performance between the conditions are due to success, failure, an interaction between these, or sampling error.

Seashore and Bavelas (33) found a decrement in mental age scores on the Draw-a-Man test under conditions of frustration. The subjects were children of varying ages. In a famous experiment designed to get at the relation between frustration and regression, Barker, Dembo, and Lewin (3) demonstrated a regression in mental age of young children as a result of stress.

One of the most intensive attempts to evaluate the use of stress as a psychometric device for personnel selection was undertaken by the Aviation Psychology Program of the Army Air Forces (27). With the use of various psychomotor tasks such as steadiness, aiming, etc., the effects of verbal threat and distraction upon performance were measured. Since this program was primarily concerned with prediction of success in training schools, very little systematic information concerning the direct effects of stress upon the measured performance is available. Many of the studies were conducted without adequate control groups. However, in our analysis of these data (27), the following facts seem to be revealed: (a) Where it can be established that stress produced an

effect, the effect seemed to be a small decrement in performance. (b) An adaptation-effect to the stress condition is suggested by the fact that the stress tests were the only psychomotor tests which produced different results according to their position within the batteries of tests. The scores for the stress tests were poorest when they occurred first in the battery.

In psychomotor performance, as well as in verbal performance, impairment of ability has not always been found to be the only result of stress. Lindsley (18) found that stress induced by rapid pacing caused the subjects to attempt more problems, which resulted in a greater number of errors and more variability between subjects. Total scores remained approximately the same because the increased speed was offset by an increase in the number of errors. Further data in the type of experiment described by Lindsley were obtained by McKinney *et al.* (22). Two kinds of reactions to stress were found: most subjects speeded up performance at the expense of accuracy and were more variable, while a few subjects showed a stable performance without any increase of speed or errors. As a whole, the stressed subjects showed a reduction of efficiency in performance in terms of the percentage of correct items to the number attempted. Clark and McClelland⁴ reported that subjects with mild fear of failure performed better, particularly towards the end of a test. Finally, improved performance under failure-stress was found by Gates and Rissland (9) with the use of motor coordination and color naming tasks.

The effects of stress upon the components of performance. Most of the studies of stress report findings concerning the total scores on the performance of some task. There is usually no attempt made to analyze these total scores into their various components and to observe the effects of stress upon them. Two subjects could obtain identical scores on a task as the result of very different patterns of performance. This is especially true where scores may be analyzed in terms of speed and accuracy and where an increase in speed may go along with a decrease in accuracy. A report in terms of total scores may obscure what is really happening to the subject's performance.

Only two studies to date have presented data relevant to this problem. In both of these studies an increase in errors was accompanied by an increase in speed as a result of stress (16, 18). In one of these studies, however, speed accounted for so much of the variance of the total scores

⁴ CLARK, R. A., & MCCLELLAND, D. C. A factor analytic integration of imaginative, performance and case study measures of the need for achievement. Unpublished research, cited in McClelland (19).

that any increase in errors was ineffective in changing the total score (16). It is probable that the effects of stress upon such components as speed and accuracy may be very different from task to task.

Adaptability to the experimental conditions and various other less easily measured aspects of performance may also be differentially affected by stress. One problem in this area is the difficulty of determining which measure represents the subject's efficiency in any particular task. In some situations accuracy may be of greater importance than speed, while in others rapidity of performance may be the important feature.

Qualitative observations of performance under stress. In connection with experimental studies of the effects of stress upon performance, many qualitative changes in behavior have been noted. Stereotyped responses (23), inattention (44), disorganized activity (36), and increased overt activity (8) have been some of the aspects of behavior under stress that have been observed. Many experimenters have reported signs of emotional upset such as sweating, tremor, subjective anxiety, pulse changes, etc. A whole new theory of the physiology of the organism (34) has grown up around the organic changes observed as the result of physiological and psychological stress.

Investigators have attempted to make field observations on the effects of stress upon performance by means of questionnaires, interviews, and rating scales. These studies have the merit that they attempt to discover the effects of non-laboratory stresses upon behavior. However, they have the great disadvantage of depending entirely upon self-report techniques. The most serious difficulty with these studies is that they do not include actual, objective measurements of performance. The questionnaires are always administered after the stressful situation has passed away, sometimes years later (5).

Dollard (5) administered a series of questionnaires on fear in battle to veterans of the Abraham Lincoln Brigade of the Spanish Civil War. These questionnaires suffered from too many limitations to be of much use. The techniques of sampling, question-wording, the nature of the attitudes measured, etc., make this study difficult to evaluate. The data do tell us something about the conditions which produce stress in actual combat and some of the personal reactions to these stresses. However, there is no way of knowing how the intense fears reported by Dollard's subjects affected their performance.

A somewhat more satisfactory questionnaire approach was attempted by Shaffer (35), with the use of veteran AAF flying personnel. Following an analysis of the results of his study, Shaffer (35) concluded that the adequate stimulus for fear is a highly motivated situation

towards which the individual has no adequate means of adjustment. This definition seems to fit most of the experimental studies of behavior under stress.

Shaffer reported many different signs of emotional upset in combat. Some of his data suggested that the severe stress of battle may, indeed, have a serious detrimental effect upon the various skills required in combat. Subjective feelings of fear increased with the number of missions that were flown, though the pilots questioned believed that this was probably not accompanied by an increase in the deterioration of behavior. Somewhat greater faith can be placed in the information from this report because of the cross-checks, the closer proximity in time of the interviews to the experience in question, and the somewhat more adequate control of sampling. Nevertheless, Shaffer's study still suffers from a complete dependence upon self-report.

The studies of Shaffer are the only field observations which actually attempted to assess the effects of combat stress on performance through self-report. The statement of a majority of the men questioned was that fear increased their efficiency or at worst made no difference in it. Only from 17 to 31 per cent of the men in the various groups interviewed reported that fear decreased the efficiency of their performance. Those men who reported an increased efficiency as the result of stress said that mild fear increased their efficiency more than intense fear.

The findings of this study are extremely interesting and provocative, but there is a distinct probability that the reports of the men interviewed do not reflect the actual state of their performance under stress. It is likely that the interviewees confused effort with performance. Also, under the circumstances of testing, men might be extremely unwilling to admit behavioral disorganization under combat. The individuals who were interviewed had no good criteria for an evaluation of their own performance under stress. The stressful conditions themselves probably reduced the accuracy of self-estimates of performance. As a result, we suspect that the percentage of men experiencing no detrimental effects of stress was greatly overestimated. In light of the experimental studies on performance under stress, it does seem likely that some individuals will show a facilitation of performance under such conditions, but it also seems probable that these individuals will be in the minority in any randomly chosen sample of people. This interesting problem remains to be studied by more reliable methods.

With the use of narco-analytic techniques, Grinker and Spiegel (11) studied battle-fatigued soldiers of the last war. They suggested that a primary mechanism for psychological breakdown under combat

involves strong guilt feelings and conflict over supposed inadequacy and cowardly behavior during combat. While it is provocative, this study is less relevant here because all of the subjects involved were psychiatric patients.

Personality correlates of behavior under stress. Very little information has been obtained about the relationship between various measures of personality and reaction to stress. The problem has theoretical as well as practical importance. On the one hand, while great individual differences in response to stress have been recognized, few fruitful attempts have been made to discover their nature. On the other hand, it would be most useful to be able to predict which people will be adversely affected by a stressful situation.

Most of the studies which have obtained relationships between performance under stress and measures of various aspects of personality have presented correlations which are statistically significant but are too small to be of practical value. For example, Taylor and Farber (39) found that submissive children showed a decrement in performance and an increase in variability under failure-stress, while ascendant children showed an improvement. In Hutt's (13) study, previously cited, maladjusted children showed impaired mental age scores on a stressful administration of the Stanford-Binet. Lazarus and Eriksen (16) have demonstrated that students with high grade-point averages in college tend to improve under stress while poor students show a decrement and greater variability in performance when ACE scores are held constant. The authors account for this relationship by the suggestion that some poor students may obtain poor grades because of the stressful nature of college examinations.

In a study that is difficult to interpret, Meadow (25) found that women with low dominance feelings were more emotionally disturbed by failure than women with high dominance. In this experiment emotional disturbance was measured by subjective report; the author found that women emotionally disturbed by the failure showed poorer performance in arithmetic and memory tests for emotional words than did those women who reported no emotional disturbance. It is impossible to tell to what extent the subjects' reports of disturbance may have been influenced by their actual level of performance. Individuals who showed poor performance may have attempted to rationalize this by claiming to be emotionally upset during the test. The study, however, is in accord with the results found by Taylor and Farber (39).

The only study that has reported relationships between performance under stress and personality variables of sufficient magnitude to be

of practical use is that of Williams (43). In a validation study of certain variables on the Rorschach test, Williams reported a multiple correlation coefficient of .824 between average decrement under stress and two scoring categories on the Rorschach. There are several features about this study, however, which suggest caution in accepting the results. In the first place, the number of cases upon which the correlations were based was only 25. There are strong possibilities of sampling errors with so few cases. In the second place, the material used was overlearned, so that the scores for the subjects under stress could only decrease. Thus, nothing but "decrement" scores were obtained. In addition, it is to be noted that the stress was produced partly by the distracting influence of the threat of electric shock.

The doubts concerning the generality of the results of this study are supported by negative findings in a study correlating a large series of Rorschach variables with the effects of stress upon performance (7).

Also on the negative side is a study by Adams (1), who found no relationship between neurotic tendency scores on the Bernreuter Personality Inventory and performance under failure-stress. In a similar experiment using the Bernreuter, Marquart (23) also obtained negative results.

McKinney *et al.* (22) found no relationship between attitudes and feelings expressed on a questionnaire and efficiency of performance under stress. Some slight positive relationships are suggested, however, with variability in a few Rorschach responses. However, the measure of variability of these Rorschach responses is not described in the experimental report.

Relevant to the problem of the personality correlates of performance under stress is an experiment by Taylor (38). She found that individuals who show high anxiety as measured by a questionnaire are people who learn a conditioned eyelid response most rapidly. Since the conditioned eyelid response is of the avoidance type, the author concludes that this superiority is the result of a high anxiety drive necessary for rapid avoidance conditioning.

Performance under stress as a predictor. There have been two systematic attempts to use performance under stress as a predictor of success in other dangerous and stressful activity. One was the well-known OSS study, and the other was an attempt by the Aviation Psychology Program to design stress tests.

The more elaborate of these experiments was the program conducted by the Office of Strategic Services to select men who would be most

effective in intelligence operations in enemy territory during World War II (46). The basic procedure was to subject a group of applicants to a series of stress tests designed by the psychological staff of the OSS. The test situations were highly realistic in the sense that they did a good job of simulating stressful situations to which the candidate was liable to be subjected while on duty. The men were aware that they were being evaluated. The stress that was created was basically a failure-stress. All measurements of performance were based upon ratings by continually observing psychologists. While large individual differences were found by the raters, these were not successfully related to later performance in the field. A possible reason for this was that the various subjects in this experimental program served in different theaters of war and performed different functions, so that the criterion of performance was not comparable. The OSS study cannot be considered to have achieved successful prediction of performance under stress, despite its elaborate design.

The Aviation Psychology Program (27) made an attempt to relate performance on a series of psychomotor tests given under stressful conditions to success in pilot training, bombardier training, and navigator training. No significant relationships were found between any one of the five tests used and any available measure of success in training school. Several major criticisms of this study may be given. First, the criteria used were poor ones inasmuch as they were very heavily saturated with semi-academic abilities. Stress may actually lower the validity of a test when the criterion is success in school. Moreover, performance in training school may be poorly related to performance under operational conditions that involve stress. In support of this last statement is a study of the validity of selection and classification tests in the theaters of war. The conclusion was drawn that many of the most important features of combat performance were not measured by the traditional abilities tests (17). Second, the most important factors measured by these AAF stress tests may not be ability to perform well under stress but the specific motor skills called for by the task itself. No attempt was made to measure deterioration under stress, or any other variable that would yield a picture of the effects of stress independent of the individual's ability. This is a problem of experimental design that we have discussed earlier.

It should be mentioned also that attempts to use the selection and classification battery of the AAF to predict the syndrome known as "anxiety-reaction" or "operational fatigue" also resulted in failure (42).

AN EVALUATION OF THE PROBLEM OF THE EFFECTS OF STRESS

The stress experiments mentioned earlier have yielded varying results: some show only a decrement in performance, others show improvement, and still others produce both of these effects for different individuals (an increase in variability). It is important to understand why the results have been seemingly inconsistent.

The distressing fact is that few of the experiments are comparable because they employed different kinds of conditions to produce stress and different kinds of tasks. The question of the dissimilarity of the two large classes of stressful conditions, failure-stress and task-induced stress, has already been raised. No data are available at present that allow us to make any generalization about this question.

Another important problem concerning the comparability of the various experiments cited is that of individual differences and sampling. Various investigators have used children, adults, women, men, college students, and military personnel. Inasmuch as large individual differences have been found in all of the experiments, it is reasonable to suppose that some of the disagreements between these experiments are due to differences in the samples employed. All in all, the body of experimental literature on the topic of the effects of stress on performance is completely unsystematic.

An integrated theoretical picture about the effects of stress upon performance must take account of individual differences, the finding of impairment as well as improvement of performance, the influence of different situations, and the effects of different kinds and amounts of stress. All of the psychological concepts about stress have been motivational in frame of reference. For convenience, we may categorize the explanatory concepts into those which emphasize primarily the energizing and directive aspects of motivation and those which emphasize the emotional aspects of motivation.

The energizing aspect of motivation. Miller (28) and Wickert (42) have emphasized fear as a motivation or drive. Since an increase in motivation is usually accompanied by an increased output in performance, Miller and Wickert both suggest that fear, produced in a stressful situation, may actually be beneficial to performance. They cite as evidence the fact that AAF personnel reported that under the stress of combat they performed more efficiently (42).

This simple notion of fear as a drive may well account for the increased efficiency that one finds occasionally for some individuals under stressful conditions. Few people would argue with the proposition that motivation does increase the adequacy of performance. It is apparent

that fear produced by a stressful situation has a considerable motivational component.

The difficulty arises because of the fact that, in some instances, high degrees of motivation or fear seem to produce a decrement or impairment of performance. Neither Miller nor Wickert takes this into account. It is therefore necessary to look for mechanisms that may supplement the notion of the simple, energizing function of motivation. It is possible to think of a critical point in the amount of fear, beyond which disruption occurs.

The directive aspects of motivation. In addition to producing the impetus to action and the persistence of action, motivation has a directive aspect and a terminating effect (26). This means that an individual will direct his efforts towards whatever operations tend to be satisfying. When the motives have been satisfied, the activity in this direction will be terminated. In extremely powerful motivational states, it is very possible that a subject's activity will be unadaptively directed. For example, Lindsley (18) and Lazarus and Eriksen (16) showed that one of the effects of a threat of failure was an increase in the speed of performance as well as an increase in the number of errors. It is quite possible that a considerable portion of the decrements in performance reported in other experiments may be due to an inappropriate increase in speed of performance.

Another illustration of unadaptive behavior as the result of stress comes from an experiment by Patrick (29), who found irrational and repetitive behavior in human subjects who were placed in a problem box and subjected to severe stress.

In many of the experiments in which failure-stress was used, it is possible that telling an intelligent subject that he has done poorly will force him to alter his mode of attack, so that it may be less effective. This kind of an effect could account, at least in part, for impairment found in some of the experimental studies.

An additional way in which the directive aspects of motivation could produce a decrement in performance is the arousal, by threat, of motives that are antithetical to the activities required by the task itself. For example, Rosenzweig (32) suggests that reactions to frustration may be need-persistent or ego-defensive. Need-persistence means that the individual will center his attention and efforts upon the frustrated need (goal-oriented behavior), and ego-defense means that the individual will be primarily concerned with the maintenance of self-esteem.

There are many instances in which activity directed toward the maintenance of self-esteem may make the effective continuance of

performance difficult or impossible. Some writers have referred to this kind of situation as stimulating the individual to leave the field psychologically. The ego-defensive aspects of the situation may become so important to the individual that he ceases or reduces his attack on the task at hand. For example, a subject in an experiment may stop work so that, even if he fails in the eyes of the experimenter, he can justify his failure to himself on the grounds that he really didn't try. The result of this defensive mechanism is most certainly to reduce the person's effort level toward the experimental task.

Thus there are other motives that may interfere with the individual's need to work at and do his best on the task at hand. In situations that threaten physical danger, the motivation to escape may directly interfere with the individual's performance. His continued performance may be, or seem to be, incompatible with self-preservation. A conflict may be produced between the desire for self-preservation and the desire to perform well. This conflict could result in impairment of the individual's level of performance.

The emotional aspects of motivation. Emotional reactions usually accompany very powerful motivation. There is no need to elaborate upon the autonomic components of emotion. The disruptive aspect of emotion, of which the autonomic components are perhaps the most characteristic, may produce mental blocks, tremors, severe anxiety-reactions that make the satisfactory completion of a given task difficult or impossible. Anxiety-reactions may include such effects as nausea, fainting, spasms, weakness, headaches, dissociation, etc. Clinicians have long been aware that the ability of people to retain lists of digits may be severely impaired by the presence of strong anxiety. This recognition finds expression in the diagnostic use of the digit-span test of the Wechsler-Bellevue Intelligence Scale.

One of the effects of anxiety might be to produce a powerful distraction. Threatened subjects frequently report that their productive thinking is disrupted by the compelling preoccupation with the thought of the consequences of failure or danger. We might consider that in some tasks, e.g., those that require fairly automatic responses, this preoccupation would have little effect, whereas in others, e.g., those that require concentration, this preoccupation would be very disrupting.

In the simplest sense, the autonomic overflow in strong emotion would make physical performance difficult or impossible. The individual who collapses before going into combat is protected against the dangers of combat. The person who develops writer's cramp is temporarily removed from the threatening examination. It is difficult to

get adequate subjective reports on the milder effects of autonomic disturbance that may interfere with an individual's performance, because subjects may magnify these effects in order to make use of them as ego-defensive mechanisms. It is possible, for example, that mild autonomic reactions may be accompanied by an improvement in performance, even though the individual may insist that he is too nervous to perform efficiently.

The cause of these disruptive influences probably lies directly with the fear itself, though the impairment may be enhanced by the presence of conflict between the anxiety produced by the threat of danger and the necessity of remaining in the situation and seeing the task through. The disruption may also be increased by the inability of individuals to cope with the fear-producing stimuli. It is possible that situations in which the individual can remove or reduce the stimuli that threaten him may not be so stressful.

Interaction of emotion and motivation. When we try to analyze what is happening in any individual's performance under stress, it is usually almost impossible to separate the effects due to the operation of emotional disruption from those due to the directive characteristics of motivation. Some subjects in an experimental situation may show impairment in performance because of ego-defensive reactions. Some may suffer primarily because of emotional disturbance. Others may be facilitated. Still other cases may involve the operation of both facilitating and disrupting motivational and emotional reactions. In fact these two may balance in any individual to produce no effect upon any measure of performance.

These interactions present a major problem for the prediction of reactions to stress. The same effect upon performance may be achieved via several different routes. Some of the negative findings with respect to the relationships between personality variables and performance under stress may be due to the assumption by many psychologists that impairment is solely related to the lack of emotional control. Because of the multiplicity of causes for any individual's particular reaction to stress, it is not surprising that the scores on inventories of neuroticism and other measures of emotional stability show negligible correlations with performance under stress.

Motivation, emotion, and the kind of stress-situation. We have pointed out that the type of motivation, that is, ego-defensive or need-persistent, the kind of approach a subject makes to the situation, and the effects of strong emotion are all possible bases for an individual's response to the stress situation. It is necessary to point out, however, that the

importance of any of these factors will depend, to a large extent, upon the conditions of the situation and the type of stress involved, as much as upon the personality and past experience of the individual. In other words, ego-defensive reactions are apt to be especially prominent in situations that utilize failure as the source of the stress. On the other hand, distraction situations, which are not accompanied by threats of failure or disparagement, are less apt to make ego-defensive reactions necessary.

Essentially, this boils down to a consideration of interactions between persons and types of stress. It would be interesting to know what kind of individual develops anxiety reactions to task-induced stress. We might guess that such people are highly motivated to perform well. The successful understanding of any individual's performance under stress depends upon some way of measuring the kinds and strength of his motivations and relating them to the characteristics of the situation in which he must perform. The fulfillment of this aim is, indeed, no simple affair.

Task components and the interactions of emotion and motivation. As we have said earlier, relatively little attention has been paid to the study of the components of the total performance of persons in stressful situations. What is going on, in terms of the effects on the subject's performance, may be masked by attention to only the total score. This score is simply the end product of his attack on the problem, his degree and direction of effort, the effects of emotional disturbance, and his abilities.

A consideration of the various components of performance may well yield a clue as to the bases of the observed effects of stress upon performance. For example, periodic blocking in a continuous task could indicate some degree of emotional disturbance in one subject. These blockings might adversely affect speed and regularity and, therefore, result in an impaired performance. On the other hand, in the same situation, another subject might show a greatly increased rate of performance at the expense of accuracy, which would suggest that the energizing and directing aspects of motivation are involved. Both subjects end up with the same total score, for different reasons, and attention to the components of that score would provide some leads about what those reasons might be.

Relating the individual components of performance to possible causes of the stress effects would not be simple. In many instances the relationship may be ambiguous. For example, excess overt activity which has occasionally been reported as an effect of stress would be

difficult to assign to any of these possible causes. Impairment might reflect emotional blocking or misguided effort as a result of false failure-information. Some help would be obtained, however, by studying the combination of effects, both qualitative and quantitative. Despite the difficulties inherent in the approach we have been suggesting, an attack on the source of the various stress effects and how they operate must be made before we can begin to understand the bases of the effects of stress upon skilled performance. It is our belief that the various individual components of performance can be more successfully related to personality variables than the total score.

Implications for future research. The most obvious questions that require an answer are:

1. How do the effects of psychological stress vary with the nature of the task?
2. What are the differences in effects produced by failure-stress and task-induced stress?
3. How does psychological stress affect the various individual components of total performance?
4. What is the relationship between personality variables, past history, and performance under stress?
5. What is the relation between the capacity of the subject and reaction to stress?
6. What relation do emotion and motivation have to performance under stress?

Research in the area of psychological stress must begin practically afresh. Any systematic program must take into account the difficulty of producing realistic stress situations and making effective measurements of the stress effects which are independent of the skills required by the task itself. It seems certain that a really systematic attack on the problems of psychological stress will produce some valuable answers to both theoretical and practical problems in behavior.

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